8 A radio is connected to the mains supply using a step-down transformer.

(a) Draw a labelled diagram of the structure of a basic step-down transformer.

[3]

(b) Explain the operation of a basic transformer.

.....

.....

.....

..... [3]

(c) The voltage of the mains supply is 230 V. The output voltage of the transformer is 6.0 V.

Calculate the value of the turns ratio $(\frac{N_s}{N_p})$. Give your answer to two significant figures.

value of turns ratio = ... [2]

[Total: 8]